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Problem Solving or Responsibility Avoidance? The Role of CEO Internal Attribution Tendency in Shaping Corporate Downsizing in Response to Performance Shortfalls

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ABSTRACT This study examines how CEOs' internal attribution tendency – to attribute an observed outcome to internal factors – shapes the extent to which corporate downsizing activities are driven by performance shortfalls. Despite the known effectiveness of downsizing in creating leaner organizations and enhancing firm performance, CEOs often avoid it as it runs counter to their own interests. Applying the awareness-motivation-capability (AMC) framework, we argue that CEOs' internal attribution tendency influences their awareness of responsibility for performance shortfalls, and ultimately whether they engage in downsizing actions in response to such shortfalls. This effect becomes more salient when CEOs have the motivation (e.g., scrutiny from financial analysts) or capability (e.g., the existence of an unfavourable external environment) to avoid taking such responsibility. Using a sample of 2424 US firms, we find support for our arguments. Our study highlights the importance of CEO internal attribution tendency in explaining whether firms resort to downsizing actions to address performance shortfalls.

Keywords: internal attribution tendency, heuristics and biases, performance feedback, strategic leadership, corporate downsizing

INTRODUCTION

Research on performance feedback suggests that when performance falls short of aspirations, firms engage in problemistic searches for a solution (Bromiley, 2004; Cyert and March, 1963). The tenets of performance feedback have been used to explain why firms facing a performance shortfall undertake organizational changes (for a review, see Greve and Gaba, 2020), including growth-related strategies such as mergers and

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acquisitions (M&As) (Iyer and Miller, 2008; Kim et al., 2015), factory expansion (Audia and Greve, 2006), and new product introductions (Gaba and Joseph, 2013; Tyler and Caner, 2016), or conversely, corporate downsizing activities (Shimizu, 2007). However, this stream of literature provides inconsistent findings on how proactively firms respond. While some scholars suggest that performance shortfalls trigger proactive searches for organizational adaptation solutions, others argue that problemistic search and organizational change do not always happen, and even that performance shortfalls may activate organizational rigidity (for a review, see Posen et al., 2018).

Such inconsistency largely arises from the implicit assumption that all top managers are fully aware of their responsibility for firm performance (Gopinath and Hoffman, 1995), and problemistic search is therefore considered to be an automatic and natural response to performance shortfalls.[1] This is contradicted by the upper echelons research suggesting that top managers exhibit significant heterogeneity in cognition and personality (Hambrick and Mason, 1984; Hitt and Tyler, 1991). As boundedly rational decision-makers are subject to cognitive limits and biases (Cyert and March, 1963; March and Simon, 1958), their awareness of their responsibility for firm outcomes may differ, resulting in heterogeneous responses to performance shortfalls.

Our study focuses on a fundamental cognitive characteristic that shapes CEOs' awareness of their responsibility for performance shortfalls: internal attribution tendency, which captures the extent to which a CEO attributes an observed outcome to internal factors (Heider, 1958; Kelley, 1973; Weiner, 1985). Where a CEO's internal attribution tendency is strong, the CEO is more likely to see the firm's performance as his or her personal responsibility and thus to respond to performance feedback. However, where a CEO's internal attribution tendency is weak, the CEO is more inclined to attribute performance to factors beyond his or her control and hence less likely to undertake actions to address performance issues.

To investigate the role of internal attribution tendency in shaping strategic actions in response to performance shortfalls, we focus on corporate downsizing – a corporate restructuring activity aimed at reducing personnel and streamlining organizations to increase resource allocation efficiency (Nixon et al., 2004). Corporate downsizing provides an ideal context in which to examine CEOs' awareness of their responsibility for firm outcomes because, while it can be a solution to adversity for firms, it is not generally aligned with CEOs' personal interests. That is, although downsizing may improve a firm's short-term performance by releasing resources and enhancing operational efficiency (Brauer and Laamanen, 2014; Johnson, 1996; Pearce and Robbins, 1994), reducing operational scale and laying off employees tend to be perceived negatively by external stakeholders and reflect poorly on the CEO, whose pay and social status are often tied to firm size (Agarwal, 1981; Chiu et al., 2020; Jensen, 1986). This is particularly true when a performance shortfall occurs; downsizing may be perceived as an admission of poor strategic decisions in the past (Brauer, 2006; Johnson, 1996) and thus goes against the desire to preserve a positive self-concept (Sedikides and Strube, 1997).

Leveraging the awareness-motivation-capability (AMC) framework from competitive dynamics research (Chen, 1996; Chen et al., 2007), we posit that CEOs with a stronger internal attribution tendency will be more *aware* of their responsibility for performance outcomes, and hence are more likely to take responsibility for performance shortfalls and

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less likely to eschew downsizing to address them, even if downsizing may be perceived negatively by stakeholders. Conversely, those with a weaker internal attribution tendency will be less likely to accept responsibility for poor performance and are thus less likely to engage in downsizing activities.

Taking a step further, we posit that the role of internal attribution tendency in shaping the relationship between performance shortfalls and downsizing actions depends on CEOs' *motivation* and *capability* to avoid responsibility for performance shortfalls. When the motivation to avoid taking responsibility (e.g., when analysts' scrutiny of the firm is intense) or the capability to do so (e.g., by blaming an unfavourable external environment) is stronger, the influence of internal attribution tendency on the relationship between performance shortfalls and downsizing actions will be stronger. Using a sample of 3977 CEOs of 2424 US listed firms between 2002 and 2013, we develop a measure of CEO internal attribution tendency based on quarterly earnings conference call transcripts and find support for our theory.

Our study contributes to research on organizational responses to performance shortfalls by highlighting the influence of CEOs' internal attribution tendency. Despite the potential of corporate downsizing to improve short-term performance, CEOs may shun it for fear of its being negatively perceived by external stakeholders and potentially linked to a reduction in compensation and social status (Shi et al., 2017b). Our work extends recent research on executive attributes and performance feedback (e.g., Blagoeva et al., 2020; Lee et al., 2022; Lim and Audia, 2020; Schumacher et al., 2020) by introducing the construct of internal attribution tendency and highlighting its importance in influencing the adoption of downsizing activities to address performance shortfalls. In doing so, our study also extends research on managerial heuristics and biases. A large body of related work has investigated how various executive heuristics and cognitive biases (e.g., self-serving bias, narcissism, hubris, and overconfidence) can play a role in strategic decision-making (e.g., Chatterjee and Hambrick, 2007; Foss and Weber, 2016; Hodgkinson and Healey, 2011; Li and Tang, 2010; Nadkarni and Chen, 2014). Extending this stream of research, our study introduces the construct of executive internal attribution tendency and highlights its importance in shaping firms' strategic decisions under performance shortfalls.

In addition, our study extends the AMC framework used in competitive dynamics (Chen and Miller, 2012; Chen et al., 2007; Downing et al., 2019) to performance feedback research. Our findings suggest that executives' *awareness* of their responsibility for firm performance will influence their response to a performance shortfall, which is also contingent on whether they have the *motivation* and *capability* to avoid taking responsibility for performance shortfalls. Lastly, our article contributes to the corporate restructuring literature. Even though downsizing is known to free up resources and improve efficiency, CEOs are less keen to undertake downsizing than growth-oriented activities such as business expansion or acquisitions (Chen, 2015; Dial and Murphy, 1995), since downsizing is often negatively perceived and may undermine their image and their interests (Chiu et al., 2020). We advance this line of research by introducing a new predictor – CEO internal attribution tendency – for corporate downsizing activities.

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THEORETICAL BACKGROUND

CEO Internal Attribution Tendency

According to attribution theory, people have an innate tendency to make sense of their surroundings. Like 'naive psychologists' (Heider, 1958), they seek to determine the cause of events in an effort to establish control over their lives and improve their ability to predict future events (Kelley, 1971). Attribution theory suggests that in answering 'why' questions, people distinguish between internal (self) and external (outside of self) explanations, thereby determining the locus of causality (Heider, 1958; Kelley, 1967). Identifying the locus of causality has been at the heart of attribution theory since its inception (Martinko et al., 2006). When the cause is seen as related to an attribute of the individual such as his or her effort or ability, it is referred to as 'internal'; when the cause is seen as related to situational factors beyond the individual's control, it is referred to as 'external'.

Building on social psychology research, management scholars have investigated how different types of executive attribution (i.e., internal or external) influence corporate leaders' decisions (e.g., Lee et al., 2022; Park et al., 2021; Shi et al., 2019a; Shin et al., 2022). However, although internal or external attribution can be situational (e.g., situational on positive or negative outcomes),[2] an individual is likely to be dominated by either a stable internal or external attribution tendency, and this tendency triggers distinct responses in the individual when problems arise (Calder and Burnkrant, 1977; Neumann, 2000; Ryan and Deci, 2002; Turban et al., 2007). Executives with an internal attribution tendency attribute performance outcomes to internal factors and accept responsibility for them, whereas those who attribute performance outcomes to external factors avoid taking responsibility.

Our study focuses on CEOs' internal attribution tendency – to what extent CEOs draw internal (versus external) inferences about their responsibility for an outcome. Those with a strong internal attribution tendency accept responsibility for an observed outcome, be it favourable (positive performance) or unfavourable (poor performance). Those with a weak internal attribution tendency will point to external factors, accepting neither responsibility nor credit for the outcome. Internal attribution tendency – an important cognitive characteristic – is particularly relevant in the context of performance feedback research because it shapes awareness of responsibility for performance outcomes,[3] which in turn influences whether a firm engages in problemistic search and the choice of strategic actions to address performance issues.

Performance Feedback and Corporate Downsizing

Management scholars have made a significant effort to understand the role of performance feedback in organizational behaviour. The behavioural theory of the firm holds that performance feedback prompts organizational adaptation and that decision-makers assess the need to change existing routines by benchmarking performance against an *aspiration level* – a satisfactory performance level (Cyert and March, 1963) – below which they may resort to *problemistic search*, defined as a search for solutions, with the goal of reversing the decline and improving future performance (Greve, 1998, 2003b). Scholars have

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examined a wide range of growth-oriented decisions triggered by performance feedback, including spending on research and development (R&D) and innovation (Chen and Miller, 2007; Giachetti and Lampel, 2010; Greve, 2003a; Tyler and Caner, 2016; Vissa et al., 2010), strategic alliances (Baum et al., 2005; Schwab and Miner, 2008; Shipilov et al., 2011), acquisitions (Haleblian et al., 2006; Iyer and Miller, 2008; Kim et al., 2015), new product introduction (Gaba and Joseph, 2013), and internationalization (Jung and Bansal, 2009). Yet despite the theory's appeal – that growth-oriented strategies are pursued to address performance shortfalls – the empirical evidence is mixed (Greve, 2008; Iyer and Miller, 2008).

Accordingly, scholars have started to explore mechanisms that may indirectly affect organizational responses to performance feedback, such as organizational design, governance structures, or external factors that stop organizations from pursuing change (Audia and Greve, 2006; Desai, 2016; Gaba and Joseph, 2013; Greve, 2008; Joseph and Gaba, 2015). Recent work has looked at the role of top executive characteristics in shaping organizational responses to performance feedback in terms of growth-oriented risk-taking activities. For example, Schumacher et al. (2020) found that CEO overconfidence is associated with a lower willingness to increase risk-taking in response to negative performance feedback, but a stronger inclination to decrease risk in response to positive feedback. Blagoeva et al. (2020) suggested that a firm's response to inconsistent performance feedback is contingent on the CEO's power – specifically that firms with a powerful CEO reduce investment in R&D as a response to inconsistent feedback.

In response to a performance shortfall, in addition to growth activities that improve their market position and performance, firms may resort to corporate restructuring, including divestiture, spin-offs and sell-offs, downsizing, and reorganization.[4] Our study focuses on downsizing, which is conducted to reduce personnel and the scale of the firm (Nixon et al., 2004). Compared with other forms of restructuring, corporate downsizing, especially workforce reduction, is regarded as a quicker, simpler way to respond to performance shortfalls and is therefore widely used to address performance issues (Cameron, 1994). By releasing resources and enhancing operational efficiency, downsizing is also more effective in improving financial performance than other restructuring activities (Brauer and Laamanen, 2014; Lee and Madhavan, 2010; Love and Nohria, 2005). It is therefore a valid strategy in the presence of a performance shortfall (Shimizu, 2007).

However, downsizing can adversely impact a CEO in two ways. First, it is associated with reducing operational scope and laying off employees, which may be perceived negatively by external stakeholders and provoke resistance from employees (Chiu et al., 2020). Second, it reduces firm size, and since CEO compensation is often a function of size (Agarwal, 1981; Jensen, 1986), it may adversely influence CEO compensation. Thus, CEOs may eschew downsizing activities as a means of addressing poor performance (Johnson, 1996).

In the next section, we hypothesize that CEOs' internal attribution tendency can significantly influence their responses to performance shortfalls by undertaking downsizing actions.

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HYPOTHESES

To develop hypotheses about the role of CEOs' internal attribution tendency in responding to performance shortfalls, we turn to the AMC framework (Chen, 1996). Building on social cognition and organizational change (Fiske and Taylor, 2013; Kiesler and Sproull, 1982), the AMC framework was originally developed in the field of competitive dynamics (Chen and Miller, 2012; Downing et al., 2019) to show how three independent components – awareness, motivation, and capability – trigger responses and actions by competitors. However, since the AMC framework by nature pertains to a response process from 'problem sensing' to 'problem solving' (Kiesler and Sproull, 1982), it has great potential to be applied to other fields related to generalized responses to problems. Recent research has applied the framework to other contexts such as firms' responses to the threat of shareholder activism (Shi et al., 2020), the threat of negative earnings surprises (Guo et al., 2020), and interorganizational spillover (Shi et al., 2022). These studies contend that the central cognitive aspect of awareness pertains to 'problem sensing'; how efficiently 'problem sensing' translates into 'problem solving' will depend on the individual's motivation and capability (Shi et al., 2020, 2022). We extend the scope of the AMC framework to managerial decisions in response to performance shortfalls by positing that internal attribution tendency shapes the CEO's *awareness* of responsibility for such shortfalls and exploring how the degree to which such tendencies shape strategic response hinges on a CEO's *motivation* and *capability* to avoid responsibility for performance outcomes.

CEO Internal Attribution Tendency: Awareness of Responsibility for Performance Shortfalls

In the AMC framework, awareness is considered 'a prerequisite for any move' in the decision-making process (Chen, 1996, p. 110). The psychology literature proposes that internal attribution tendency is closely related to self-awareness (Duval and Silvia, 2002), i.e., 'making internal attributions of responsibility for events that have negative consequences causes people to feel more aware of themselves' (Rogers et al., 1982, p. 959). Thus, a CEO's internal attribution tendency can influence their *awareness* of responsibility for firm performance, shaping whether the firm resorts to corporate downsizing activities in response to performance shortfalls.

We posit that CEOs with a strong internal attribution tendency will attribute observed outcomes to internal factors (i.e., their own responsibility for poor performance). Accordingly, they are more willing to learn from past experiences (Desai, 2015) and proactively take strategic actions to address performance shortfalls. They perceive that organizational actions (e.g., strategic decisions taken, sales efforts, product or process development, or steps that impact production efficiency) affect firm performance (Heider, 1958; Kelley, 1973; Weiner, 1985). As a result, to address performance shortfalls, they will be likely to undertake downsizing activities even if these run counter to their own interests in the short run. In contrast, those with a weak internal attribution tendency will blame external factors (e.g., the economy, the competition, or regulators) for poor firm performance (Ford and Baucus, 1987; Milliken and Lant, 1991). They will especially care about the negative impact of downsizing actions on themselves and

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emphasize that there is little they can do to improve performance. Since they perceive downsizing as sending a signal about having chosen poor growth strategies in the past (Capron et al., 2001), they will eschew such activities to avoid being blamed for poor performance. Thus, we propose:

Hypothesis 1: The stronger a CEO's internal attribution tendency, the stronger the impact of a firm's performance shortfall on its downsizing actions in the subsequent period.

Moderating Effects of CEOs' Motivation and Capability to Avoid Responsibility

We next investigate how the *motivation* to avoid taking responsibility for firm performance influences CEOs' internal attribution tendency in shaping firms' responses to performance shortfalls through downsizing. In the AMC framework, *awareness*, characterized as individuals' cognition of taking actions, is the first stage of the response process. Motivation, which captures the subjective motive to actively respond, comes in a close second in the psychological drive to take action to respond (Chen, 1996). We investigate the role of one source of motivation – scrutiny by financial analysts – because of analysts' important role as external evaluators who can influence CEOs' motivation to take responsibility for performance shortfalls.

Research in a wide range of academic disciplines has considered how managers operate under the watchful eye of financial analysts (Benner and Ranganathan, 2012; Gentry and Shen, 2013). Financial analysts evaluate CEO performance by raising questions about the firm's operations, collecting proprietary data via meetings and conference calls, and disseminating that information to the public via reports (Lang et al., 2004). They have been shown to influence managerial compensation and dismissal decisions (Chen et al., 2015; Wiersema and Zhang, 2011). This could provide motivation for CEOs to ingratiate themselves with financial analysts (Westphal and Clement, 2008).

When firms are covered by many financial analysts, CEOs come under the scrutiny of these external performance evaluators and thus have strong motivation to project a favourable image. Corporate downsizing in response to a performance shortfall may signal that they have made unwise strategic decisions in the past, such that CEOs with a weak internal attribution tendency will be reluctant to engage in downsizing to address such shortfalls. In contrast, when analyst coverage is sparse, CEOs are less subject to external scrutiny or pressure, and hence those with a weak internal attribution tendency may feel less concerned about undertaking downsizing activities that could otherwise be associated with notions of incompetence or poor strategic decisions. Where the internal attribution tendency is strong, CEOs are aware of their responsibility for poor performance and therefore decisions are less likely to be influenced by how they will be perceived by financial analysts. Consequently, they are more likely to resort to downsizing in response to shortfalls whether or not the firm is closely followed by analysts. It follows that the moderating effect of a CEO's internal attribution tendency will be stronger in the presence of a higher level of analyst coverage.

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Hypothesis 2: The positive moderating effect of CEO internal attribution tendency on the relationship between performance shortfalls and downsizing actions is stronger when analyst coverage is higher.

We now discuss how the *capability* to avoid responsibility for firm performance affects the role of internal attribution tendency with respect to downsizing in response to performance shortfalls. The third component of AMC refers to the capability to respond and act. Unlike *motivation*, which implies a subjective attitude to taking action, *capability* largely depends on objective dimensions such as resource endowment or the external environment (Chen, 1996). In this respect, we consider the moderating effect of an unfavourable/favourable external environment in developing our third hypothesis.

The external environment is closely related to a CEO's capability to avoid responsibility for firm performance. A less favourable external environment, such as an economic crisis, makes it easier for CEOs with a weak internal attribution tendency to blame external factors for poor performance (Eggers and Song, 2015; Martinko et al., 2006). Put differently, when they can avoid responsibility for performance shortfalls, they refrain from corporate downsizing activities that harm their interests. In contrast, if the external environment is favourable, it is difficult for CEOs with a weak internal attribution tendency to ascribe poor performance to external factors and deny their own responsibility; as a result, these CEOs may be more willing to engage in downsizing activities to address performance shortfalls. We posit that for CEOs with a strong internal attribution tendency, strategic decisions are less influenced by the external environment – be it favourable or unfavourable – since they accept responsibility for firm performance and thus will undertake corporate downsizing to address performance shortfalls. In contrast, CEOs with a weak internal attribution tendency react differentially in favourable/unfavourable external environments. In this sense, internal attribution tendency plays a more salient role in moderating the effect of performance shortfalls on corporate downsizing actions when the external environment is unfavourable.

Hypothesis 3: The positive moderating effect of CEO internal attribution tendency on the relationship between performance shortfalls and downsizing actions is stronger when the external environment is more unfavourable.

METHOD

Data and Sample

Our sample selection reviewed all US firms covered by the Thomson Reuters StreetEvents database from 2002 to 2013. StreetEvents provides complete coverage of conference call transcripts of 6367 firms during our sample period and has been used in recent management research to capture executive characteristics (e.g., DesJardine and Shi, 2021; Shi and DesJardine, 2022). We used quarterly earnings conference call transcripts from this database to measure a CEO's internal attribution tendency. Prior

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research has used letters to shareholders in annual reports to capture top managers' causal attributions (e.g., Bettman and Weitz, 1983; Clapham and Schwenk, 1991; Lee et al., 2004; Schwenk, 1993; Staw et al., 1983). Compared with these letters, CEOs' speeches during earnings conference calls more directly reflect their cognitive style (Bowen et al., 2002). To test our hypotheses, we merged StreetEvents with other databases, including Compustat, CRSP, BoardEx, and Thomson Reuters' 13F database, to generate the final sample of 3977 CEOs, 2424 firms, and 17,727 firm-year observations.

Predictors

Performance shortfalls. Various performance indicators have been used to measure a firm's performance relative to its aspiration levels, such as return on assets (ROA) (Audia and Greve, 2006; Chen and Miller, 2007; Shimizu, 2007) and market share (Baum et al., 2005). We chose ROA as it is widely used in this line of research (Shinkle, 2012).

Managers use organizational historical performance and performance of peer firms to measure aspiration levels (Cyert and March, 1963). A historical performance shortfall is based on a recursive measure that weights prior performance and prior aspiration levels to measure current aspiration levels (Kuusela et al., 2017). It is defined as $A_t = \alpha P_{t-1} + (1-\alpha)A_{t-1}$, where A_t refers to the aspiration level at time t, and P_{t-1} is the firm performance at time $t-1$. The coefficient **alpha (α)** indicates the relative importance of the prior aspiration level $(t-1)$ versus the actual prior performance $(t-1)$. Following Kuusela et al. (2017), we chose a relatively large value of α (=0.75), but the results are robust, α = 0.6 or 0.5.

The social aspiration level is typically measured by comparing the focal firm's performance with the industry average/median (Audia and Greve, 2006; Baum et al., 2005; Iyer and Miller, 2008). Since CEOs do not pay the same attention to all players in the same industry, preferring to focus on a smaller reference group of firms that are similar in size and belong to the same strategic group as their peers (Fiegenbaum and Thomas, 1995), we construct our social aspiration levels by using a matching approach to identify the most relevant peer firms. Specifically, we use the Text-based Network Industry Classification (TNIC-3) to identify peer firms (Hoberg and Phillips, 2016).[5] We apply exact matching based on year and TNIC-3 industry classifications to identify the n closest matching firms for each observation, relying on a product similarity score and total assets. Then we choose the peer firms by ranking the Mahalanobis distance score between the focal firm and industry peers (Mahalanobis, 1936). Given that CEOs have cognitive limitations and are more likely to focus on a smaller number of comparable industry peers, we follow Kuusela et al. (2017) to set n = 5. To construct our social aspiration level, we use the average performance of matched peers.

We then combine social and historical aspiration to create an overall aspiration level $(= a \times \text{Social aspiration} + (1-a) \times \text{Historical aspiration})$. We follow Harris and Bromiley (2007) to give equal weights to social and historical aspirations (α = 0.5).[6] Moreover, we follow prior research to split the performance measure into two variables. The first variable is calculated as performance minus the aspiration (social and historical

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aspiration), but only for those observations where performance is *below* aspiration, and the variable is set to zero when performance is above the aspiration. This means that the variable's values are negative when performance is below the aspiration, with lower (more negative) values indicating poorer performance. This variable is coded as *ROA-aspiration* (*below*). Our theory focuses on performance below aspiration (performance shortfall), and we take the *absolute values* of performance shortfalls to facilitate interpretation – the greater the value, the higher the performance shortfall.

The second variable from the split performance measure, coded as *ROA-aspiration* (*above*), equals performance minus aspiration (social and historical) when performance is *above* the aspiration and is set to zero when performance is below aspirations. Lower (higher) values indicate poorer (better) performance. This is used as a control variable in our main analysis.

CEO internal attribution tendency. As noted, we used CEOs' speeches during quarterly earnings conference calls to measure CEO attribution. Conference call transcripts have two parts: prepared remarks and question and answer (Q&A) sections. Our readings of transcripts suggest that CEOs make attribution statements in both. We therefore used both prepared remarks and Q&A sections to measure CEO attributions. We followed guidance on computer-aided textual analyses (McKenny et al., 2018; Short et al., 2010) to analyse CEOs' words in these transcripts.

We first extracted CEOs' speeches from earnings conference call transcripts and identified causal sentences using causal words included in the Linguistic Inquiry and Word Count (LIWC) text analysis program (Pennebaker et al., 2007). Based on these causal sentences, we captured CEOs' internal attribution and external attribution. As psychology research suggests that personal pronouns reflect the attention allocation of the speaker (Davis and Brock, 1975; Gunsch et al., 2000; Tausczik and Pennebaker, 2010), we assume that CEOs exhibit strong internal attribution when they frequently use first-person pronouns (both *I* and *we* forms) in causal attribution sentences. We thus counted how frequently a CEO used first-person pronouns to capture *internal* attributions. As managers often refer to general economic and external factors when making external attributions (Baginski et al., 2004), to construct our word list for external attributions, we randomly selected 100 transcripts and read them to identify external factor words.[7] We counted how frequently a CEO used external factor words to measure *external* attributions. Thereafter, *CEO internal attribution tendency* is measured as the ratio of the number of internal attribution words used by CEOs in causal sentences in earnings calls in a year to the number of both internal and external attribution words.[8] We conducted several analyses to validate our measure (see Appendix 1).

Dependent Variable

Our dependent variable is the number of *downsizing actions*. Downsizing happens when a firm phases out a product line, closes facilities, or reduces its workforce. We obtained downsizing data from the Capital IQ Key Development database. Capital IQ aggregates information from more than 20,000 news sources, regulatory filings, and company websites to capture important business decisions made by firms, and it has been used in

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recent management and accounting research (Jin et al., 2012; Livnat and Zhang, 2012; Pan et al., 2015; Shi et al., 2018). References to downsizing actions included the following examples: '*American Airlines to reduce flights*,' '*Honeywell International to cut jobs across its Phoenix-based Aerospace Division*,' and '*Alcoa to close operations in Mexico and Texas*.' We counted the number of downsizing actions announced by the firm in a year to measure our dependent variable. A total of 6610 downsizing actions were announced for firms in our sample during the years under investigation. Around 82 per cent of firm-year observations were associated with no downsizing action, 10 per cent with one downsizing action, and the remaining 8 per cent with more than one downsizing action.

Moderators

The first moderator is *analyst coverage*, which is measured as the number of financial analysts covering a firm in a year (Wiersema and Zhang, 2011). We took the natural log of analyst coverage, as the moderating effect of analyst coverage may decrease as the value of analyst coverage increases. The data used to measure analyst coverage were obtained from the International Brokers' Estimate System (IBES) database. The second moderator is *unfavourable external environment*. We use the 2007–08 financial crisis to capture an unfavourable external environment (Erkens et al., 2012). The crisis was triggered by excessive risk-taking by banks and led to the most severe global recession since the Great Depression (Erkens et al., 2012), which created an unfavourable external operating environment for companies. This moderator takes a value of 1 for years 2007 and 2008, and 0 for other years. Our results are similar if we use only year 2008 to denote the financial crisis.

Control Variables

As indicated earlier, our theoretical focus is performance below aspiration (i.e., performance shortfall); therefore, we first control for performance above aspiration (*ROA-aspiration* [*above*]) in all models. We also include several firm- and CEO-level control variables that may influence organizational change. At the firm level, we first control for *firm size* (measured by taking the natural log of total assets) as larger firms may have more assets to be downsized. We also control for slack resources. 'Slack' refers to the excess resources available to an organization during a given planning cycle (Nohria and Gulati, 1996), which may affect firms' willingness to engage in downsizing actions. We follow prior research to create a composite slack index, *slack resources*, by adding the standardized scores of three categories: available slack (measured by current assets divided by current liabilities), absorbed slack (measured by working capital divided by total sales), and potential slack (measured by the equity-to-debt ratio) (Bromiley, 1991; Chen, 2008). Data used to measure firm size and slack were obtained from Compustat.

We next control for firm *product diversification*, which consists of related diversification and unrelated diversification. 'Unrelated' refers to diversification arising from business units that operate in different 2-digit SIC industry groups, and 'related' refers to diversification arising from business units that operate in the same 2-digit SIC industry group (at the 4-digit SIC industry level) (Hoskisson et al., 1993). We use an entropy measure based

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on segment sales revenues to capture unrelated and related diversification separately and then add them to measure product diversification. We do not use the TNIC-3 industry classification to measure product diversification because it cannot be matched to segment sales data from Compustat. We next control for firms' engagement in *other restructuring actions* instead of downsizing actions as a response to performance shortfalls. Thus, we control for the sum of a firm's announced actions on business reorganizations (e.g., '*Kraft Foods announced to reorganize its European operations*'), on plans to sell/divest (e.g., '*Liberty Media puts On Command unit on sale*'), and on plans for spin-offs/split-offs (e.g., '*US Bancorp plans to spin off Piper Jaffray companies*').

We also control for *institutional ownership concentration*, measured by the Herfindahl–Hirschman Index of institutional ownership, as prior research (Chaganti and Damanpour, 1991) suggests that ownership concentration level can affect corporate strategies. Although *analyst coverage* is one of our moderators, we control for it because pressure from analysts can lead firms to pursue corporate downsizing actions (Zuckerman, 2000).

At the CEO level, we control for *CEO gender* (taking a value of 1 if a CEO is male, and 0 if female) and *CEO age*, based on data from BoardEx. We also control for CEO incentive scheme using *CEO option pay ratio*, the ratio of the value of all the options held by the CEO to the value of annual compensation (Sanders, 2001; Sanders and Hambrick, 2007). We also control for *CEO overconfidence*, as overconfident CEOs may be less willing to engage in downsizing actions. We follow prior research to measure overconfidence by examining the extent to which the CEO retains in-the-money options that are vested (Banerjee et al., 2015; Malmendier and Tate, 2008); the higher the value, the more overconfident the CEO. Data used to measure CEO option pay and overconfidence were obtained from Capital IQ Compensation.

Powerful CEOs may have more discretion to resist downsizing actions. We followed prior research to create a power index, *CEO power*, which includes the following indicators: CEO duality (scored as 1 if a CEO is also the board chair, and 0 otherwise), CEO tenure (the number of years in the CEO position in the focal firm), CEO equity ownership (the ratio of the number of shares owned by the CEO to the total number of shares outstanding), board independence (the number of outside directors divided by the board size; the ratio is reversely coded to indicate higher CEO power), and percentage of CEO-appointed directors (the number of directors appointed by CEOs divided by the board size). These five indicators capture different sources of CEO power (Finkelstein and Hambrick, 1990; Haynes and Hillman, 2010; Westphal and Zajac, 1995; Zhu and Chen, 2015), and we standardize and sum them to measure *CEO power*. Data on CEO duality, CEO tenure, board independence, and percentage of CEO appointed directors were collected from BoardEx, and data on CEO ownership, from Thomson Reuters Insiders.

A firm's downsizing actions can be influenced by peers' downsizing actions because of imitation among industry peers (Abrahamson and Rosenkopf, 1993). Therefore, we control for *industry downsizing actions* by using the average number of downsizing actions conducted by industry peers (based on the TNIC-3 industry classification). Lastly, we control for year fixed effects and industry fixed effects using two-digit SIC codes. Table I reports descriptive statistics and correlations for variables used in this study.

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Table I. Descriptive statistics and correlations

	Variable	*Mean*	*SD*	*Min*	*Max*	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*
1	Downsizing actions	0.56	1.75	0.00	54.00	1.00																
2	CEO internal attribution tendency	0.66	0.13	−1.00	1.00	0.04	1.00															
3	Financial crisis	0.20	0.40	0.00	1.00	0.09	0.02	1.00														
4	Analyst coverage (log)	1.95	0.90	0.00	4.02	0.17	0.01	−0.03	1.00													
5	ROA-aspiration (below)	0.04	0.06	0.00	0.91	0.01	0.01	0.00	−0.13	1.00												
6	ROA-aspiration (above)	0.02	0.06	0.00	1.81	−0.04	0.01	0.02	−0.05	−0.19	1.00											
7	Firm size (log)	7.10	1.94	2.30	12.15	0.32	0.09	−0.01	0.60	−0.21	−0.21	1.00										
8	Slack resources	−0.06	2.33	−2.42	18.11	−0.10	−0.07	−0.01	−0.13	0.05	0.21	−0.43	1.00									
9	Firm diversification	0.48	0.56	0.00	1.91	0.16	0.01	−0.01	0.10	−0.03	−0.11	0.31	−0.18	1.00								
10	Other restructuring actions	0.10	0.30	0.00	3.05	0.39	0.07	0.02	0.15	0.04	−0.05	0.27	−0.10	0.15	1.00							
11	Institutional ownership concentration	0.09	0.14	0.00	1.00	−0.06	0.02	0.00	−0.18	0.02	0.00	−0.21	0.03	−0.16	−0.06	1.00						
12	CEO age	62.04	7.63	34.00	97.00	0.04	0.03	0.02	−0.05	−0.03	−0.02	0.08	−0.05	0.10	0.02	−0.04	1.00					
13	CEO gender	0.98	0.15	0.00	1.00	−0.01	0.00	−0.01	0.01	−0.03	−0.02	0.03	0.00	0.01	0.00	0.00	0.03	1.00				
14	CEO overconfidence	0.28	0.23	0.00	0.92	−0.09	0.00	0.00	0.04	−0.09	0.05	−0.06	0.05	−0.04	−0.08	−0.06	0.03	0.01	1.00			
15	CEO option pay percentage	2.60	5.85	0.00	39.79	−0.02	−0.06	−0.08	0.13	−0.07	0.07	0.04	0.06	−0.05	−0.02	−0.08	0.06	0.01	0.47	1.00		
16	CEO power	0.64	1.96	−2.77	8.53	−0.10	−0.07	0.00	−0.13	0.02	0.03	−0.23	0.08	−0.05	−0.09	0.03	0.19	0.03	0.06	0.02	1.00	
17	Industry downsizing actions	0.50	0.38	0.00	7.00	0.14	−0.01	0.13	−0.01	0.06	−0.03	0.03	−0.06	0.08	0.09	−0.02	0.07	−0.03	−0.11	−0.03	−0.01	1.00

Note: N = 17,727. Absolute value of correlations greater than 0.03 statistically significant at $p < 0.05$ level at both tails.

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Models

The dependent variable of our study is the number of downsizing actions, which is a count variable. The two commonly used methods to analyse count data are negative binomial regression and Poisson regression. Given that our dependent variable has the property of overdispersion, we use negative binomial regressions to test our hypotheses. To address serial correlation in panel data sets, we adopt random-effects negative binomial models. Such models treat the data in a time-series cross-section panel and can handle the spatial and temporal effects in the data. In all regressions, we measure our dependent variable at $t+1$ and all the other variables at t.

RESULTS

Table II presents results used to test Hypothesis 1. In Model 1, the coefficient estimate of *ROA-aspiration* (*below*) is positive and statistically significant ($\beta = 1.21$, $p < 0.01$), suggesting that performance shortfall is positively associated with the number of downsizing actions. Hypothesis 1 suggests that a firm's performance shortfall is positively associated with the number of downsizing actions when the CEO has a strong internal attribution tendency. In Model 2, the coefficient estimate of *CEO internal attribution tendency* × *ROA-aspiration* (*below*) is positive and statistically significant ($\beta = 10.84$, $p < 0.01$), supporting Hypothesis 1. In terms of magnitude, when CEO internal attribution tendency takes a low value (mean − 1 SD), the number of downsizing actions *decreases* by 11 per cent of its mean when performance shortfall increases from a low value (mean − 1 SD) to a high value (mean + 1 SD). However, in the presence of high CEO internal attribution tendency (mean + 1 SD), the number of downsizing actions *increases* by 56 per cent of its mean for the same increase in performance shortfall.

Scholars caution against the interpretation of interaction terms in non-linear models such as negative binomial regressions. We follow Wiersema and Bowen (2009) to calculate marginal effects of performance shortfalls (using the 'margins' command in Stata) on the number of downsizing actions when CEO internal attribution tendency takes different values. As shown in Table III, the marginal effect of *performance shortfall* is negative for a lower level of *CEO internal attribution tendency* ($\beta = -0.48$, n.s.) and positive for a higher level of *CEO internal attribution tendency* ($\beta = 2.45$, $p < 0.01$), consistent with Hypothesis 1.

Hypothesis 2 proposes that the moderating effect of CEO internal attribution tendency on the relationship between performance shortfalls and downsizing actions will be stronger when there is a higher level of analyst coverage. Model 3 of Table II is used to test this hypothesis. The coefficient estimate of *CEO internal attribution tendency* × *ROA-historical aspiration* (*below*) × *Analyst coverage* is 4.58 ($p < 0.05$). To test Hypothesis 2, we present marginal effects of performance shortfalls in Table IV when CEO internal attribution tendency and analyst coverage take on different values. When analyst coverage takes a low value, the difference in the marginal effect of performance shortfalls on downsizing actions in the presence of low and high CEO internal attribution tendency is 1.9 (= 2.97 − 1.07); when analyst coverage takes a high value, the difference in the marginal effect of performance shortfalls on downsizing actions in the presence of low

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Table II. Random effects negative binomial regressions: moderating effect of CEO internal attribution tendency

Variables	*Model 1*	*Model 2*	*Model 3*	*Model 4*	*Model 5*
CEO internal attribution tendency × ROA-aspiration (below)		10.84**	1.99	7.22*	0.00
		[2.53]	[4.33]	[3.00]	[4.27]
CEO internal attribution tendency × ROA-aspiration (below) × Analyst coverage			4.58*		3.99†
			[2.23]		[2.16]
CEO internal attribution tendency × ROA-aspiration (below) × Unfavourable external environment				11.99*	10.48*
				[4.94]	[4.83]
CEO internal attribution tendency	−0.00	0.00	0.61	0.34†	0.98*
	[0.17]	[0.17]	[0.41]	[0.19]	[0.43]
ROA-aspiration (below)	1.21**	0.96*	3.68**	1.38**	4.08**
	[0.38]	[0.40]	[0.63]	[0.45]	[0.66]
CEO internal attribution tendency × Analyst coverage			−0.28		−0.29†
			[0.17]		[0.17]
ROA-aspiration (below) × Analyst coverage			−1.52**		−1.49**
			[0.32]		[0.32]
CEO internal attribution tendency × Unfavourable external environment				−1.31**	−1.29**
				[0.31]	[0.31]
ROA-aspiration (below) × Unfavourable external environment				−1.41*	−1.47*
				[0.71]	[0.68]
Unfavourable external environment				−0.01	−0.01
				[0.08]	[0.08]
ROA-aspiration (above)	0.75†	0.74†	0.66	0.70†	0.63
	[0.42]	[0.42]	[0.43]	[0.43]	[0.43]
Firm size	0.48**	0.48**	0.48**	0.48**	0.48**
	[0.02]	[0.02]	[0.02]	[0.02]	[0.02]
Slack resources	−0.00	−0.00	−0.00	−0.01	−0.00
	[0.01]	[0.01]	[0.01]	[0.01]	[0.01]
Firm diversification	0.01	0.01	0.02	0.01	0.01
	[0.04]	[0.04]	[0.04]	[0.04]	[0.04]
Other restructuring actions	0.22**	0.22**	0.23**	0.22**	0.23**
	[0.04]	[0.04]	[0.04]	[0.04]	[0.04]
Institutional ownership concentration	0.12	0.13	0.11	0.10	0.08
	[0.18]	[0.18]	[0.18]	[0.18]	[0.18]
Analyst coverage	−0.01	−0.01	0.00	−0.01	−0.00
	[0.03]	[0.03]	[0.03]	[0.03]	[0.03]
CEO age	−0.01	−0.01†	−0.01†	−0.01	−0.01†
	[0.00]	[0.00]	[0.00]	[0.00]	[0.00]

(Continues)

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Table II. (Continued)

Variables	*Model 1*	*Model 2*	*Model 3*	*Model 4*	*Model 5*
CEO gender	−0.40**	−0.40**	−0.40**	−0.42**	−0.42**
	[0.12]	[0.12]	[0.12]	[0.12]	[0.12]
CEO overconfidence	−0.63**	−0.62**	−0.62**	−0.62**	−0.62**
	[0.10]	[0.10]	[0.10]	[0.10]	[0.10]
CEO option pay ratio	−0.01**	−0.01**	−0.01**	−0.01**	−0.01**
	[0.00]	[0.00]	[0.00]	[0.00]	[0.00]
CEO power	−0.05**	−0.04**	−0.04**	−0.05**	−0.05**
	[0.01]	[0.01]	[0.01]	[0.01]	[0.01]
Industry downsizing actions	0.22**	0.22**	0.22**	0.23**	0.23**
	[0.05]	[0.05]	[0.05]	[0.05]	[0.05]
Constant	−3.34**	−3.33**	−3.34**	−3.34**	−3.34**
	[0.69]	[0.69]	[0.69]	[0.69]	[0.69]
Observations	17,727	17,727	17,727	17,727	17,727
Industry FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Log-likelihood	−10,023	−10,013	−10,002	−10,000	−9989
Chi-squared	2459	2469	2498	2488	2520

Note: Robust standard errors in brackets.
**p < 0.01; *p < 0.05; †p < 0.1. Two-tailed tests.

Table III. Marginal effect of performance shortfall on downsizing actions

CEO internal attribution tendency	*Marginal effect*	*Z-statistics*	*p-value*	*95% confidence interval*
Low	−0.48	−0.81	0.42	[−1.65, 0.68]
High	2.45	5.58	0.00	[1.58, 3.30]

Note: Low CEO internal attribution tendency refers to mean minus one standard deviation, and high CEO internal attribution tendency refers to mean plus one standard deviation.

and high CEO internal attribution tendency is 3.87 (= 1.54 − [−2.33]). These findings suggest that the moderating effect of CEO internal attribution tendency is stronger in the presence of high analyst coverage than in the presence of low analyst coverage, consistent with Hypothesis 2.

Hypothesis 3 proposes that the moderating effect of CEO internal attribution tendency on the relationship between performance shortfalls and downsizing actions will be stronger when the external environment is unfavourable. Model 4 of Table II is used to test this hypothesis. The coefficient estimate of *CEO internal attribution tendency* × *ROA-historical*

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Table IV. Marginal effect of performance shortfall on downsizing actions conditional on analyst coverage

CEO internal attribution tendency	*Analyst coverage*	*Marginal effect*	*Z-statistics*	*p-value*	*95% confidence interval*
Low	Low	1.07	1.77	0.08	[−0.12, 2.26]
High	Low	2.97	6.36	0.00	[2.05, 3.89]
Low	High	−2.33	−2.96	0.00	[−3.88, −0.79]
High	High	1.54	2.64	0.01	[0.49, 2.69]

Note: Low CEO internal attribution tendency (analyst coverage) refers to mean minus one standard deviation, and high CEO internal attribution tendency (analyst coverage) refers to mean plus one standard deviation.

Table V. Marginal effect of performance shortfall on downsizing actions conditional on external environment

CEO internal attribution tendency	*Unfavourable external environment*	*Marginal effect*	*Z-statistics*	*p-value*	*95% confidence interval*
Low	No	−0.08	−0.11	0.91	[−1.44, 1.29]
High	No	1.99	3.51	0.00	[0.88, 3.10]
Low	Yes	−2.65	−2.68	0.01	[−4.59, −0.71]
High	Yes	2.55	4.13	0.00	[1.33, 3.76]

Note: Low CEO internal attribution tendency refers to mean minus one standard deviation, and high CEO internal attribution tendency refers to mean plus one standard deviation.

aspiration (below) × *Unfavourable external environment* is 11.99 ($p < 0.05$). Following our prior approach, we provide results in Table V, where we calculate marginal effects of performance shortfall on the number of downsizing actions when CEO internal attribution tendency and unfavourable external environment take on different values. When the external environment is not unfavourable, the difference in the marginal effect of performance shortfalls on downsizing actions in the presence of low and high CEO internal attribution tendency is 2.07 (= 1.99 − [−0.08]); when the external environment is unfavourable, the difference in the marginal effect of performance shortfalls on downsizing actions in the presence of low and high CEO internal attribution tendency is 5.20 (= 2.55 − [−2.65]). These findings suggest that the moderating effect of CEO internal attribution tendency is stronger in an unfavourable external environment. Model 5 reports the full model with all the interaction terms.

In Appendix 2, we conduct a set of robustness checks, including addressing selection bias, checking robustness of our key measures, and applying different model specifications. These analyses continue to support our theoretical arguments.

DISCUSSION

Our study examines the role of CEO internal attribution tendency in shaping whether firms engage in downsizing activities in response to performance shortfalls. Building

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on the AMC framework, we develop three research hypotheses pertaining to CEOs' awareness of their responsibility for performance shortfalls and their motivation and capability not to take such responsibility. We find that CEOs with a strong internal attribution tendency will increase the number of downsizing actions in the presence of performance shortfalls, whereas those with a weak internal attribution tendency will decrease the number of downsizing actions in the presence of such shortfalls. Furthermore, the moderating effect of CEO internal attribution tendency is stronger when firms are faced with intensive scrutiny by financial analysts and an unfavourable external environment.

Theoretical Contributions

Our findings contribute to several streams of research. First, our study adds to performance feedback research by incorporating CEOs' internal attribution tendency into the literature. Although corporate downsizing provides an effective way for firms to enhance short-term performance, CEOs may not necessarily resort to such actions to mitigate performance shortfalls. Our study suggests that whether CEOs engage in downsizing activities to address performance shortfalls hinges on the degree of their internal attribution tendency. Those with a strong internal attribution tendency are aware of their responsibility for performance shortfalls and willing to take the blame, and hence resort to any activities that can enhance firm performance, including downsizing actions, despite their potentially negative implications for the CEO. Those with a weaker internal attribution tendency are more likely to eschew downsizing activities in this situation since they fail to accept their responsibility for performance shortfalls. In essence, our findings suggest that CEOs' internal attribution tendency is critical to understand whether performance shortfalls will trigger corporate downsizing activities.

Relatedly, recent research has started to examine the role of executive characteristics such as CEO power and overconfidence in shaping firms' responses to performance feedback in terms of risk-taking propensity, resource allocations, alliances, and acquisitions (e.g., Blagoeva et al., 2020; Lee et al., 2022; Lim and Audia, 2020; Schumacher et al., 2020). Our study adds to this stream by considering the role of a CEO's internal attribution tendency in shaping downsizing actions that can lead to a reduction in firm size rather than growth-oriented risk-taking decisions.

More broadly, our study provides new insights into research on executive heuristics and biases (e.g., Chatterjee and Hambrick, 2007; Foss and Weber, 2016; Hodgkinson and Healey, 2011; Li and Tang, 2010; Nadkarni and Chen, 2014). Our study not only introduces a new executive cognitive bias – CEO internal attribution tendency – to the literature but also illustrates that the role of CEO internal attribution tendency as a cognitive bias in shaping firm strategic decisions hinges on firm financial performance. Our analyses suggest that CEO internal attribution tendency affects corporate downsizing when a firm underperforms, but such an effect does not exist in the presence of desirable financial performance. This highlights the importance of considering organizational performance in unpacking when executive heuristics and biases can affect firm strategic decisions.

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In addition, we introduce the AMC framework developed from competitive dynamics research (Chen et al., 2007; Chen and Miller, 2012) to performance feedback research. Whereas this stream of literature implicitly contends that problemistic search in response to performance shortfalls is contingent on actors' motivation (e.g., managerial risk preference) and capability (e.g., presence of slack resources) (for a review, see Posen et al., 2018), it assumes no awareness component. Our findings imply that CEOs' internal attribution tendency shapes their awareness of responsibility for performance shortfalls and how the firm responds. The effect of that awareness is dependent on their motivation (when under external scrutiny) and their capability to avoid responsibility for performance shortfalls (when the external environment can be blamed). By using the AMC framework to develop our hypotheses, we introduce a new theoretical perspective to understand what factors can shape firms' responses to performance feedback.

Our study also advances management research on executive attribution. Research has shown that CEOs are susceptible to a self-serving bias – that is, they tend to ascribe positive outcomes to themselves and negative outcomes to external factors as a means to maintain a positive self-image and/or project themselves favourably to outsiders (Hayward et al., 2004). Other research has focused on what influences boards to attribute CEO successes and failures to internal or external factors (e.g., Shi et al., 2019a). Our study departs from existing research on self-serving bias by focusing on CEOs' internal attribution tendency, a cognitive characteristic that pertains to their heterogeneous propensities for ascribing observed outcomes to internal versus external factors, which in turn may exert an influence on how firms respond to performance shortfalls.

Practical Implications

The implications of our study will be of interest to both corporate boards and executives. First, it is important for directors to understand CEOs' internal attribution tendency, notably that those with low internal attribution tendency are less willing to undertake downsizing activities in response to poor performance. Board members thus need to intensify their monitoring to ensure that CEOs have considered downsizing actions in problemistic search. Second, our study suggests that executives should make a self-assessment of their internal attribution tendency to be aware of how far it may bias their decisions and constrain their strategic options. To address this issue, firms should pair executives with respectively strong and weak internal attribution tendencies in the top management team to enhance diversity in mindset.

Limitations and Future Research Avenues

The limitations of our study – which future research could help to overcome – are as follows: First, although validity testing shows that our measure of CEO internal attribution tendency, which is based on analysing the content of CEOs' earnings conference call transcripts, is reasonably good, future research could use more fine-grained qualitative data to identify variations in internal and external attributions and examine whether different types have differential influences on organizational responses to performance

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shortfalls. For instance, although we treat CEOs' attributions to themselves *and* their organizations as internal, CEOs who attribute success or failure to themselves may differ from those who attribute these to their organizations.

Second, we focus on CEOs in our study because they are the most important decision-makers in firms. However, we have only investigated how CEOs respond to overall performance shortfalls rather than poor performance directly related to their own decision-making. Future research can extend this study by investigating whether CEOs with strong internal attribution tendency tend to engage in downsizing activities by divesting assets acquired by themselves, for which they are directly responsible, versus other assets. Moreover, large-scale investments may call for participation from other non-CEO top executives and the board of directors. An interesting avenue for future research would be to explore how attribution inconsistency between key top executives shapes strategic decisions. For example, if the CEO exhibits a high internal attribution tendency but the COO also shows a low internal attribution tendency, the influence of CEO internal attribution tendency may be attenuated given that the COO shoulders some responsibility for strategic decisions. Similarly, how the board of directors' internal attribution tendency may strengthen or weaken the CEO's internal attribution tendency offers another potential avenue for future research.

Third, we acknowledge that some CEOs may be motivated to hide their responsibility for negative outcomes even though they are aware of their responsibility for them, a behaviour triggered by self-serving attribution bias (Martinko et al., 2007; Miller and Ross, 1975). Future research could investigate what psychological attribute could trigger CEOs' hiding action, and the relationship between self-serving (or situational) attribution bias and general internal attribution tendency. In addition, research could also explore whether other psychological and cognitive attributes such as overconfidence (Malmendier and Tate, 2005), the 'Big Five' personality factors (Nadkarni and Herrmann, 2010), temporal focus (Nadkarni and Chen, 2014), and hubris (Hayward and Hambrick, 1997) influence how firms respond to performance shortfalls. For instance, those with a stronger present temporal focus are more likely to initiate organizational change as they are more sensitive to the urgency of changing performance shortfalls and engaging in intensive problemistic search.

Finally, while our study focuses on the number of downsizing actions, we acknowledge that heterogeneity in downsizing actions, such as the magnitude of downsizing and the speed of downsizing, may also be affected by performance shortfalls and CEO internal attribution tendency. Exploiting such heterogeneity, future research can explore to what extent firms tend to conduct incremental or radical downsizing. Relatedly, our study captures overall scrutiny by financial analysts using analyst coverage; however, the type of analyst recommendations (i.e., buy, hold, or sell) may also shape CEOs' motivation to avoid taking responsibility for performance shortfalls (Shi et al., 2017a). For example, when analysts issue a buy recommendation, CEOs with a weak internal attribution tendency may be more likely to avoid responsibility for poor performance. Thus, future research can investigate the role of different analyst recommendations in shaping the relationships between performance feedback, CEO internal attribution tendency, and corporate downsizing activities.

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NOTES

[1] This assumption is deeply embedded in the literature of organizational responses to performance feedback (for a review, see Kotiloglu et al., 2021).

[2] Note that our study is not related to the fundamental self-serving attribution bias, where individuals tend to attribute successes to internal factors but failures to external factors. Instead, we focus on an individual attribute: internal attribution tendency. In other words, self-serving attribution bias depends on performance, but internal attribution tendency is not situational.

[3] Awareness pertains to the extent to which CEOs are aware of their *responsibility* for firm outcomes rather than CEOs' awareness of firm performance shortfalls. It is unlikely that CEOs are not aware of such shortfalls, given their role.

[4] Corporate restructuring can be broadly classified into portfolio restructuring and organizational restructuring (Bowman et al., 1999; Bowman and Singh, 1993). Portfolio restructuring is about reducing the scope of operations, such as through divestiture or the sale of a business unit (Bergh et al., 2008). Corporate spin-offs and sell-offs that create a separate entity represent another form of restructuring (Corley and Gioia, 2004). Downsizing focuses on reducing personnel and trimming the presumed 'fat' within an organization to achieve a more efficient allocation of resources (Love and Nohria, 2005; Nixon et al., 2004). Reorganization refers to a change in organizational design, such as redrawing divisional boundaries (DeWitt, 1998) to meet the demands of a changing environment (Girod and Whittington, 2017). These four actions – divestiture, spin-offs and sell-offs, downsizing, and reorganization – are the means by which managers restructure an organization, presumably to maximize its value-creation potential (Bergh et al., 2008).

[5] The logic behind TNIC-3 is to calculate product similarity based on information disclosed on 10-K forms filed with the US Securities and Exchange Commission. When two firms have a high product similarity score, they are considered industry peers. Hoberg and Phillips (2016) explain why the TNIC-3 classification is better than the SIC classification. For instance, firms may enter and exit different industry spaces over time, making the SIC classification less reliable. In contrast, the TNIC-3 classification can mitigate measurement biases from changing industry spaces by calculating product market relatedness.

[6] Our results are largely the same if $\alpha = 0.4$, 0.6, or 0.7.

[7] Our results are similar if we include third-person pronouns in capturing external attributions.

[8] This variable is time variant. In robustness checks, we use a time-invariant measure and find consistent results.

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APPENDIX 1
VALIDATION OF THE MEASUREMENT OF CEO INTERNAL ATTRIBUTION TENDENCY

The following steps were taken to validate CEO internal attribution tendency based on content analysis. We first manually classified attributions for 200 randomly selected conference call transcripts, then identified attributions made in each of them. A statement was considered an attribution if a CEO explained the factors behind a performance outcome (be it financial, market, or operational performance). An attribution was considered internal if it focused on factors within a firm (e.g., strategic decisions or managerial efforts), and external if it referred to events outside a firm (e.g., competition, supply disruption, weather, or regulatory changes). In a very limited number of cases, statements contained both internal and external attributions.

A total of 349 attribution statements were identified among 200 conference call transcripts by one author of the study. To ensure consistency, one graduate student also independently coded the attribution statements. Inter-rater reliability between the two coders was high ($\kappa = 0.90$, $p < 0.001$) (Landis and Koch, 1977). Differences in coding were reconciled through discussion prior to validity tests. Among the 349 attribution statements, 219 were classed as internal, 111 as external, and 19 as both.

We then calculated CEO internal attribution for each attribution statement based on our dictionaries. Afterwards, we investigated the relationship between our content analysis measure and manually coded measure. For ease of presentation, we focused on the 330 (i.e., 219 + 111) coded as either internal or external. To evaluate the explanatory power of content analysis, we conducted a logistic regression with the content analysis measure as the independent variable. The dependent variable received a value of 1 if a statement was manually coded as *external*, and 0 if it was coded as *internal*. We find that the coefficient estimate of the content analysis measure (CEO internal attribution) is negative and statistically significant ($\beta = -8.46$, $p < 0.001$). The model correctly classifies 71 per cent of attributions, and the area under the receiver operating characteristic (ROC) curve is 0.72, which is considered acceptable (Hosmer and Lemeshow, 2000). These findings indicate that our dictionaries capture CEO internal versus external attributions reasonably well.

Prior research (Chatterjee and Hambrick, 2007) has measured CEO narcissism using CEOs' usage of first-person pronouns as one of the narcissism indicators, but its focus is mainly on the singular versus plural form (*I* versus *we*). Our measure of CEO internal attribution tendency includes both singular and plural forms of first-person pronouns and, moreover, differs from CEO narcissism in an important way: we focus only on sentences related to causal attributions and then use a ratio of internal attribution words to total attribution words to measure CEO internal attribution tendency. In addition, we also developed a narcissism measure for a random selected sample of 500 CEOs, following Chatterjee and Hambrick (2007). The correlation between these measures of CEO internal attribution tendency and CEO narcissism is only 0.03, alleviating the concern that our measure of CEO internal attribution tendency does not have high discriminant validity.

The list of words pertaining to internal attribution includes I, Id, I'd, I'll, Im, I'm, Ive, I've, me, mine, my, myself, lets, let's, our, ours, ourselves, us, we, we'd, we'll, we're, weve, we've.

The list of words pertaining to external attribution includes environment, demand, supplier, supply, customer, client, consumer, government, regulation, economic, economy, market, competitor, rival, competition, weather, disaster, industry.

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APPENDIX 2
ROBUSTNESS CHECKS

Selection Bias

It is possible that board members select CEOs with a stronger internal attribution tendency and simultaneously lead firms to engage in downsizing actions. In other words, there could be unobservable heterogeneity that influences CEO internal attribution tendency and our dependent variable. To address this issue, we conducted additional analyses using the inverse-probability-of-treatment-weighted (IPTW) methodology, a two-stage selection-on-observables estimation technique (Robins et al., 2000) that helps correct for the non-random assignment of CEO internal attribution tendency. IPTW has been used in prior observational research to mitigate selection bias and adjust for non-random treatment (Almandoz and Tilcsik, 2016; Shi et al., 2019b). This weighting approach is akin to what would occur if all firms had both high and low levels of CEO internal attribution tendency, addressing the possibility that specific types of firms have CEOs with a high level of this tendency.

To implement IPTW, we first estimate a probit model predicting the probability of a firm having a high level of CEO internal attribution tendency. We include all the variables used in predicting restructuring actions in addition to firm control variables and industry dummy variables. In addition, we include the *state average of CEO internal attribution tendency* as an instrument (based on where a firm is headquartered). Research suggests that firms are more likely to hire locally, such as recruiting top managers from the same state (Yonker, 2017). Thus, the internal attribution tendency of CEOs of firms located in the same state could predict that of the focal CEO. However, the former may not exert a direct influence on the focal firm's downsizing actions. In unreported probit regressions, the coefficient estimate of *state average CEO internal attribution tendency* is positive and statistically significant ($\beta = 0.669$, $p < 0.05$), consistent with our arguments.

Model 1 of Table AI reports results from IPTW regressions. In Model 1, the coefficient for *CEO internal attribution tendency* × *ROA-aspiration (below)* is positive and statistically significant ($\beta = 10.28$, $p < 0.01$), consistent with Hypothesis 1.

Table AI. Additional analyses

Variable	*Model 1*	*Model 2*	*Model 3*	*Model 4*	*Model 5*
CEO internal attribution × ROA-aspiration (below)	10.28**				
	[2.58]				
CEO internal attribution (residual) × ROA-aspiration (below)		5.69*			
		[2.62]			
CEO internal attribution (time-invariant) × ROA-aspiration (below)			10.95**		
			[2.95]		
CEO internal attribution × ROA-historical aspiration (below)				7.84**	
				[2.07]	
CEO internal attribution × ROA-social aspiration (below)					4.36**
					[1.42]
Observations	17,727	17,727	17,727	17,727	17,727
Industry FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Log-likelihood	−10,277	−10,022	−10,022	−13,463	−13,449
Chi-squared	2639	2457	2464	2729	2809

Note: Robust standard errors in brackets.
**p < 0.01; *p < 0.05. Two-tailed tests.

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Alternative Measures of CEO Internal Attribution Tendency

To alleviate the concern that our measure of CEO internal attribution tendency could be influenced by situational factors, we developed a residual measure of CEO internal attribution tendency. Specifically, we first regressed CEO internal attribution tendency on a set of predictors including firm size, ROA, Tobin's q, industry fixed effects, and year fixed effects. We then used the residual of this regression to measure CEO internal attribution tendency. Findings from this residual measure are reported in Model 2 of Table AI. The coefficient for *CEO internal attribution tendency* × *ROA-aspiration* (*below*) is positive and statistically significant ($\beta = 5.69$, $p < 0.05$). In addition, we used a time-invariant measure of CEO internal attribution tendency by taking the average of CEO internal attribution tendency for the same CEO across all the sample years. Findings using this alternative measure are reported in Model 3 of Table AI. The coefficient for *CEO internal attribution tendency* × *ROA-aspiration* (*below*) is positive and statistically significant ($\beta = 10.95$, $p < 0.01$).

Social and Historical Aspirations

Recent research (Bromiley and Harris, 2014; Washburn and Bromiley, 2012) suggests that managers respond to historical and social aspiration levels differently. We therefore also analysed historical and social performance shortfalls separately. Results from this supplementary analysis are presented in Models 4 and 5 of Table AI. The coefficients for *CEO internal attribution tendency* × *ROA-historical aspiration* (*below*) in Model 4 and *CEO internal attribution tendency* × *ROA-social aspiration* (*below*) in Model 5 are statistically significant in the hypothesized directions ($\beta = 7.84$, $p < 0.01$ in Model 4; $\beta = 4.36$, $p < 0.01$ in Model 5), suggesting that CEO internal attribution tendency moderates the influence of both historical and social performance shortfalls on downsizing actions.

Other Analyses

First, instead of testing the moderating effects of analyst coverage and unfavourable external environment using triple interactions, we also conducted subgroup analyses. To test the moderating effect of analyst coverage, we split the sample into high and low analyst coverage based on the median value of analyst coverage. In untabulated results, we find that the coefficient for *CEO internal attribution tendency* × *ROA-social aspiration* (*below*) is 14.43 ($p < 0.01$) for the high analyst coverage subgroup and 6.82 ($p < 0.05$) for the low analyst coverage subgroup. Second, to test the moderating effect of an unfavourable external environment, we split the sample into firm years during the financial crisis and those not during the financial crisis. The coefficient for *CEO internal attribution tendency* × *ROA-social aspiration* (*below*) is 13.69 ($p < 0.01$) for the financial crisis subgroup, and 5.42 ($p < 0.10$) for the non-financial crisis subgroup.

We also sum up the number of downsizing actions, business reorganizations, seeking to sell/divest, and spin-off/split-off to capture the total number of corporate restructuring actions undertaken by a firm in a year. In untabulated results, we find support for all three hypotheses using this alternative dependent variable.

We conducted firm fixed effects Poisson regressions to control for the influence of time-invariant heterogeneity on downsizing actions. Again, we find support for all three hypotheses, indicating that our results may not be biased by time-invariant firm heterogeneity. Lastly, we investigated whether CEO internal attribution tendency might influence slack search in terms of downsizing actions. We find that the coefficient for *CEO internal attribution tendency* × *ROA-aspiration* (*above*) is negative but statistically not significant.

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